Preliminary Exam 2025-26

Std : X
Time : 2 Hours

Sub. : Science & Technology (1)

Marks : 40

Q.1) A) Select the proper alternative. 5

1) ..... is an ore of aluminium.
(a) cryolite (b) Bauxite (c) Carbonate (d) Sulphate

2) A ray of light makes an angle of 50° with the surface Si of glass slab. Then its angle of incidence will be .....
(a) 50° (b) 0° (c) 90° (d) 40°

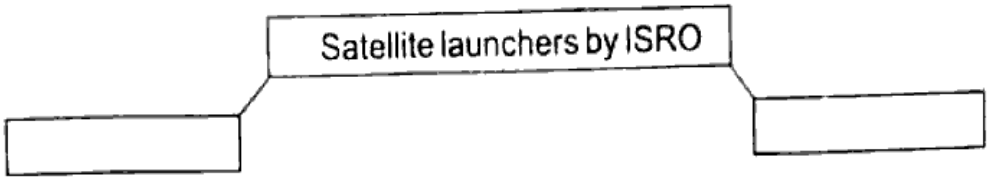
3) The reaction of iron nail with copper sulphate solution is .....
(a) Displacement (b) Double displacement (c) Combination (d) Decompositon

4) The frequency of alternate current (AC) is .....
(a) 0 Hz (b) 50 Hz (c) 60 Hz (d) 100 Hz

5) The value of persons mass on the earth surface is 80 Kg. What will be its mass on the moon?
(a) 80 Kg (b) 50 x 9.8 (c) 0 (d) None

B) Complete the concept. 5

1)



2) **Find odd man out :**
Magnet, Loud speaker, microphone, electric motor

3) **Match the columns.**

Group A	Group B
a) Alkane	i) C = C
b) Alkene	ii) C - C
	iii) C ≡ C

4) **State True or False.**
The conical cells of human eyes responds to colours.

5) **Define :** Humidity.

Q.2) a) Give scientific reason. (Any 2) 4

1) Metallic character goes on decreasing while going from left to right in modern periodic table.

2) For an electric transmission copper and aluminium wires are used.

(P.T.O)

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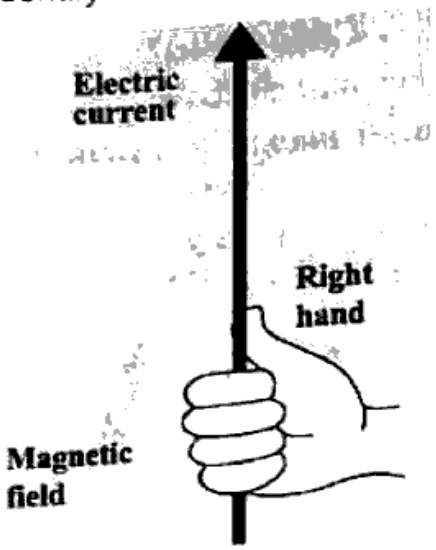
3) In old age biofocal lens is necessary for some persons.

4) In cold countries, in winter even when water of lake freezes, aquatic animals and plants can survive. 6

B) Solve any three subquestions.

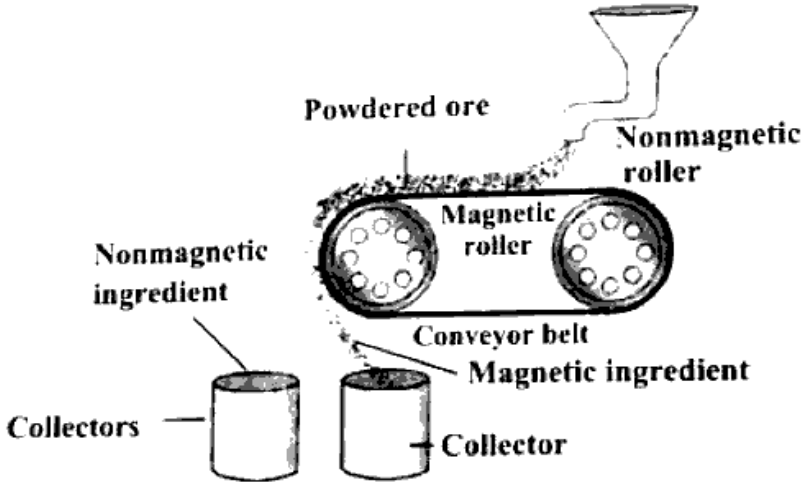
1) Explain the term.
   a) Exothermic reaction b) Endothermic reaction

2) Identify and state the rule given below.



3) An object takes 5 sec. to reach the ground from height of 5 m. on a planet. What is the value of 'g' on the planet?

4) Observe the figure and answer the questions given below.



   a) Which metal separation method is shown above?
   b) What type of metal can be seperated by using this method?

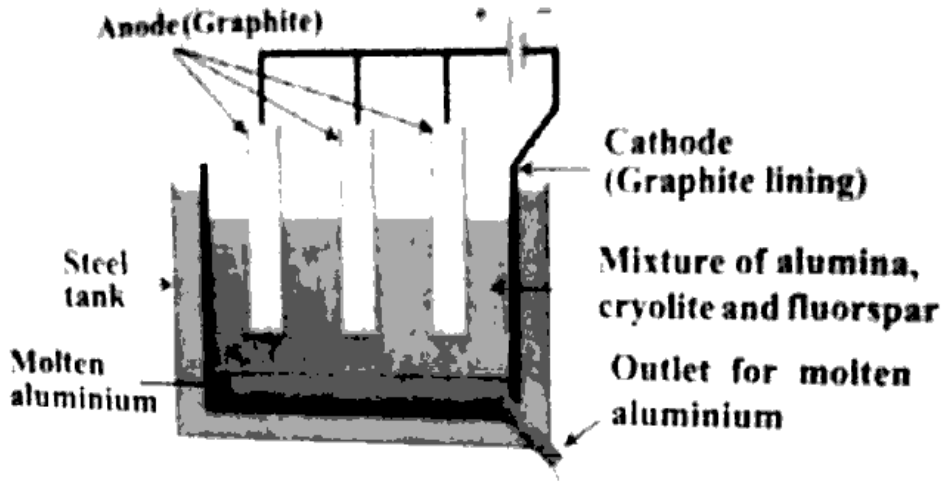
5) What is meant by artificial satellite?
   Give any 2 examples of artificial satellite made by India.

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Q. 3) Solve any five subquestions from the following.

1) Observe the figure and answer the questions given below.



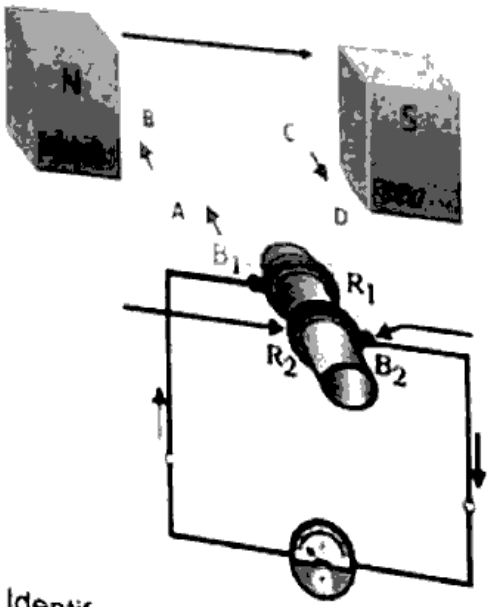
a) Which figure is given above?
b) Write molecular formula of Alumina.
c) What mixture needed to purify the metal?

2) Explain formation of rainbow with the help of its figure.

3) Explain latent heat also define latent heat of fusion and latent heat of vepourization? https://www.maharashtrastudy.com

4) Explain Newton's Universal Law of Gravitation. Give formula of it.

5) Observe the figure and answer the questions given below.



a) Identify the figure?
b) Draw and name it?
c) According to which rule does it work?

6) **Complete the table given below. (Any 1)**

Defect Name	Images formed	Corrected by lense	Possible reasons
Nearsightedness	.....	.....	....
.....	.....	Convex lens	.....

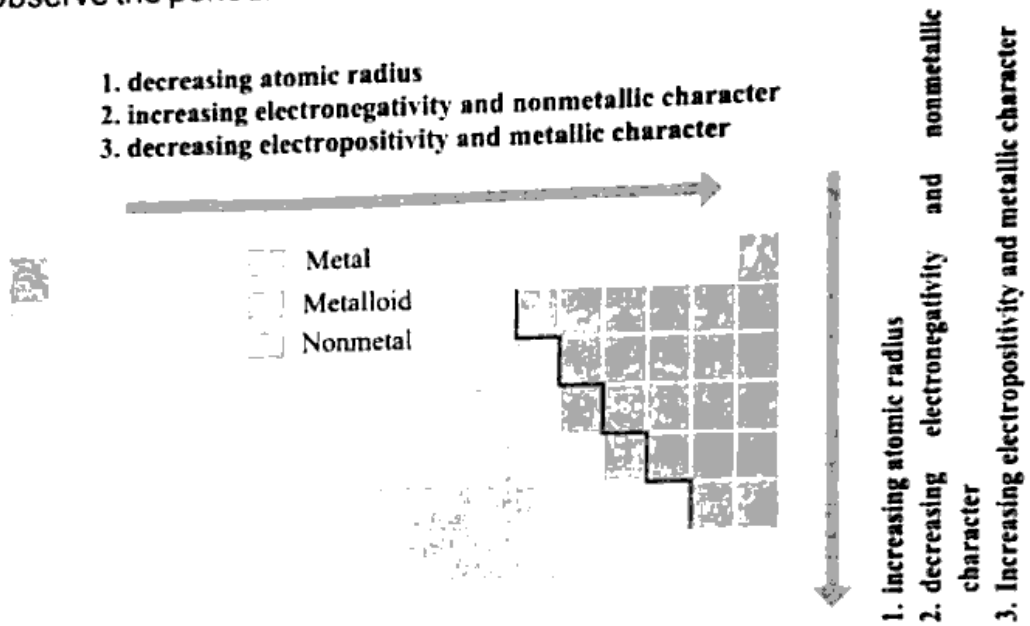
7) **Write the structural formula of following compounds.**

(a) butane (b) Propanol (c) ethene

8) Explain types of reactions according to hydrogen and oxygen. Illustrate with example. **5**

Q. 4) Solve any one subquestion from the the given questions.

1) Observe the periodic trend and answer the following questions.



(a) Metals found at which side of periodic table?
(b) What is periodic trend in atomic radius while going left to right?
(c) What trend is seen in metallic charector while down the group?
(d) Name the four blocks of modern periodic table.
(e) Group 1 and 2 are called as .....

OR

2) a) Difference between Saturated and unsaturated hydrobarbons.
b) Complete the flow chart.

$CH_4 \rightarrow$ [ ] $\rightarrow C_3\text{-}H_8 \rightarrow$ [ ] $\rightarrow$ [ ] $\rightarrow$ [ ] $\rightarrow C_7\text{-}H_{16}$

c) Define esterification?

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